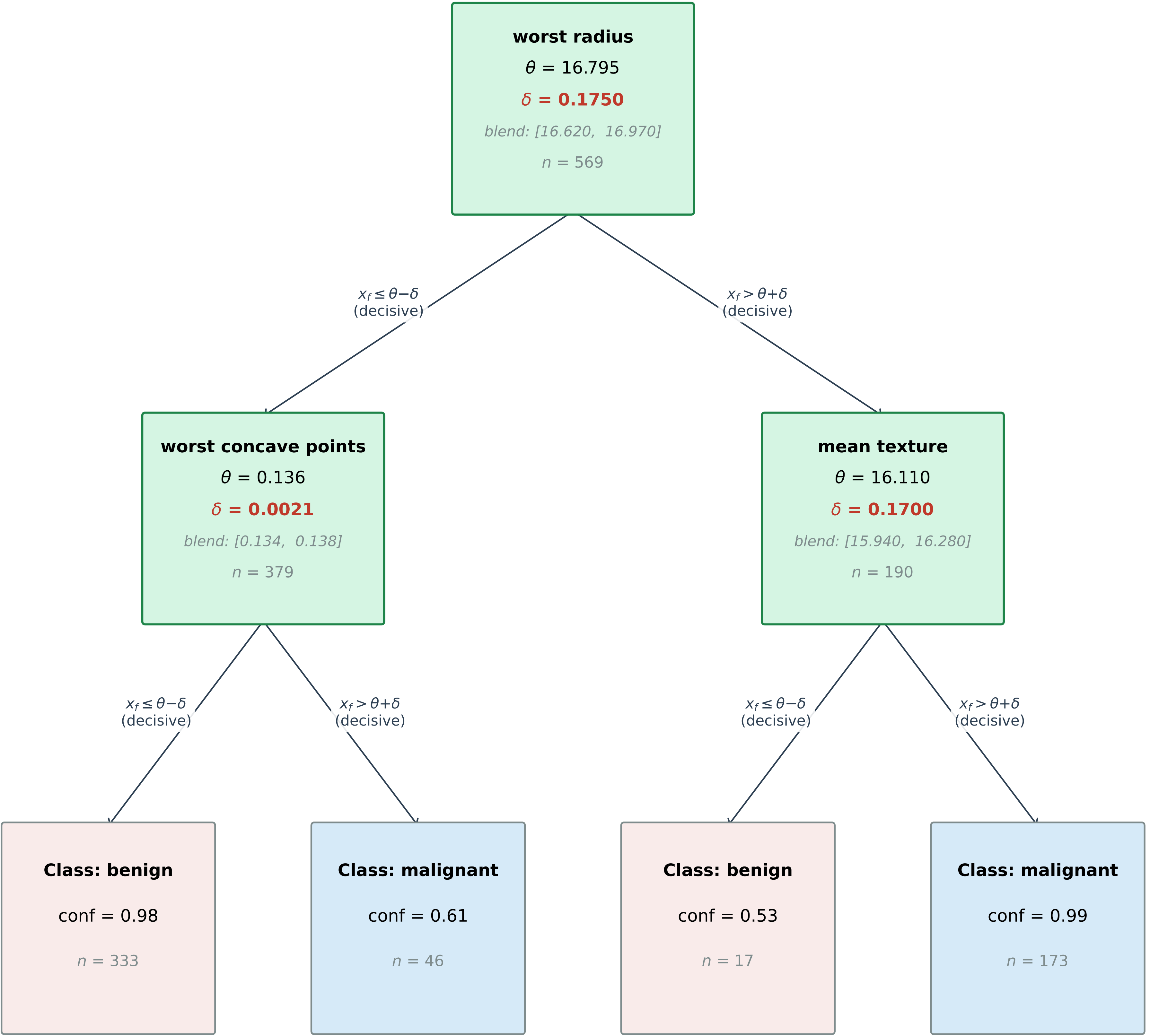


Ternary Decision Tree — Probabilistic Routing
Margin δ Method | depth 2 | Breast Cancer dataset



Probabilistic routing rules:

$x_f \leq \theta - \delta$
decisive left subtree

$\theta - \delta < x_f \leq \theta + \delta$
BOTH subtrees (blended)
 $\hat{p} = w_L \hat{p}_L + w_R \hat{p}_R$

$x_f > \theta + \delta$
decisive right subtree

Split node (θ = threshold, δ = uncertainty half-width, blend range = $[\theta - \delta, \theta + \delta]$) Leaf: benign Leaf: malignant